Problem Set 1 Walkthrough: Household Models

MSc Economics for Development - Microeconomic Theory

Francesco Losma

University of Oxford - Michaelmas Term 2026, Week 3

Today's plan

1. **Recap of Lectures 1-4:** the main ideas from the agricultural household model.
2. **PS1, Q1 (a)-(d):** a unitary household with Cobb-Douglas preferences.
3. **PS1, Q1 (e)-(f):** a collective household model.
4. **PS1, Q1 (g)-(j):** an application of the unitary household model to child labour.
5. **Wrap-up** and takeaways.

Recap: Lectures 1-4

Five things to take from the household models lectures

1. With complete markets, farm households **separate** production from consumption decisions. Production is always **efficient**. LaFave and Thomas (2016) reject the separation result. Evidence in favour of **incomplete markets**.

Five things to take from the household models lectures

1. With complete markets, farm households **separate** production from consumption decisions. Production is always **efficient**. LaFave and Thomas (2016) reject the separation result. Evidence in favour of **incomplete markets**.
2. Unitary models imply **income pooling**: only total income matters, not who earns it. Duflo (2003) finds evidence against that.

Five things to take from the household models lectures

1. With complete markets, farm households **separate** production from consumption decisions. Production is always **efficient**. LaFave and Thomas (2016) reject the separation result. Evidence in favour of **incomplete markets**.
2. Unitary models imply **income pooling**: only total income matters, not who earns it. Duflo (2003) finds evidence against that.
3. Cooperative collective models keep **efficiency**, but the allocation depends on **Pareto weights** μ.

Five things to take from the household models lectures

1. With complete markets, farm households **separate** production from consumption decisions. Production is always **efficient**. LaFave and Thomas (2016) reject the separation result. Evidence in favour of **incomplete markets**.
2. Unitary models imply **income pooling**: only total income matters, not who earns it. Duflo (2003) finds evidence against that.
3. Cooperative collective models keep **efficiency**, but the allocation depends on **Pareto weights** μ.
4. Efficiency is an assumption to test. **Commitment problems, exit options and private information** can break it.

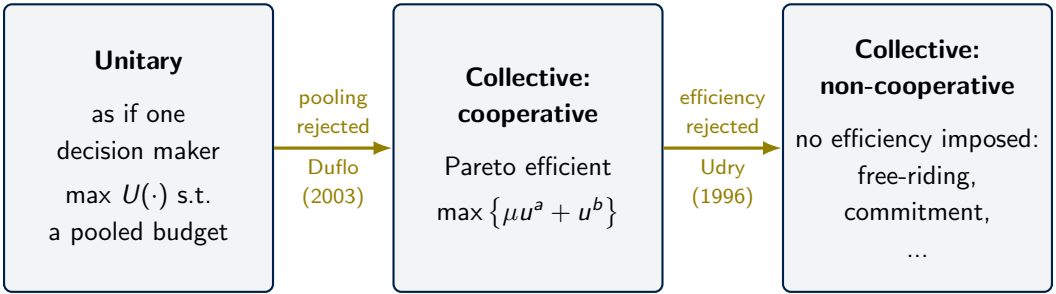
Five things to take from the household models lectures

1. With complete markets, farm households **separate** production from consumption decisions. Production is always **efficient**. LaFave and Thomas (2016) reject the separation result. Evidence in favour of **incomplete markets**.
2. Unitary models imply **income pooling**: only total income matters, not who earns it. Duflo (2003) finds evidence against that.
3. Cooperative collective models keep **efficiency**, but the allocation depends on **Pareto weights** μ.
4. Efficiency is an assumption to test. **Commitment problems, exit options and private information** can break it.
5. **Children** are consumption goods, investments and earners at once; who controls income shapes how much is invested in them.

Five things to take from the household models lectures

1. With complete markets, farm households **separate** production from consumption decisions. Production is always **efficient**. LaFave and Thomas (2016) reject the separation result. Evidence in favour of **incomplete markets**.
2. Unitary models imply **income pooling**: only total income matters, not who earns it. Duflo (2003) finds evidence against that.
3. Cooperative collective models keep **efficiency**, but the allocation depends on **Pareto weights** μ.
4. Efficiency is an assumption to test. **Commitment problems, exit options and private information** can break it.
5. **Children** are consumption goods, investments and earners at once; who controls income shapes how much is invested in them.

The map: three lenses on the household



Children: the same lenses applied to fertility, investment in kids and child labour (e.g. Qian (2008))

Toolkit: the consumer problem (used in PS1 c-d)

The problem

$$\max_{x_1,x_2} \; u(x_1,x_2) \quad \text{s.t.} \quad w \geq p_1 x_1 + p_2 x_2$$

- Non-satiation $\Rightarrow$ the budget binds.
- Lagrangian + FOCs $\Rightarrow$
  MRS = relative prices:

$$\frac{\partial u/\partial x_1}{\partial u/\partial x_2} = \frac{p_1}{p_2}$$

Toolkit: the consumer problem (used in PS1 c-d)

The problem

$$\max_{x_1,x_2} u(x_1,x_2) \quad \text{s.t.} \quad w \geq p_1 x_1 + p_2 x_2$$

- Non-satiation $\Rightarrow$ the budget binds.
- Lagrangian + FOCs $\Rightarrow$ MRS = relative prices:

$$\frac{\partial u/\partial x_1}{\partial u/\partial x_2} = \frac{p_1}{p_2}$$

Log-linear Cobb-Douglas:

$u = \alpha \log x_1 + (1-\alpha)\log x_2$

$$x_1^\star = \frac{\alpha\, w}{p_1}, \qquad x_2^\star = \frac{(1-\alpha)\, w}{p_2}$$

- Constant expenditure shares α, $1-\alpha$.
- Income elasticity $\varepsilon_i = \frac{\partial x_i^\star}{\partial w}\frac{w}{x_i^\star} = 1$.

Toolkit: the consumer problem (used in PS1 c-d)

The problem

$$\max_{x_1,x_2} u(x_1,x_2) \quad \text{s.t.} \quad w \geq p_1 x_1 + p_2 x_2$$

- Non-satiation $\Rightarrow$ the budget binds.
- Lagrangian + FOCs $\Rightarrow$ MRS = relative prices:

$$\frac{\partial u/\partial x_1}{\partial u/\partial x_2} = \frac{p_1}{p_2}$$

Log-linear Cobb-Douglas:

$u = \alpha \log x_1 + (1-\alpha)\log x_2$

$$x_1^\star = \frac{\alpha\, w}{p_1}, \qquad x_2^\star = \frac{(1-\alpha)\, w}{p_2}$$

- Constant expenditure shares α, $1-\alpha$.
- Income elasticity $\varepsilon_i = \frac{\partial x_i^\star}{\partial w}\frac{w}{x_i^\star} = 1$.

Engel Curve plots equilibrium quantity demanded as function of income $x_i^\star(w)$

- Normal good: $\frac{\partial x_i^\star}{\partial w} > 0$; inferior good: < 0.
- Luxury: $\varepsilon_i > 1$; necessity: $\varepsilon_i < 1$.

L1 - The agricultural household and the separation property

- The household **produces** $F(L, A)$ and **consumes**; with complete markets it trades labour, land and output at market prices w, r, p.

L1 - The agricultural household and the separation property

- The household **produces** $F(L, A)$ and **consumes**; with complete markets it trades labour, land and output at market prices w, r, p.
- Budget constraint:

$$\underbrace{pF(L,A) - wL - rA}_{\Pi\text{: profits of the } \textit{firm}} + \underbrace{w\left(E_1^L + E_2^L\right) + rE^A}_{\text{value of labour \& land endowments}} \geq \underbrace{p\left(c_1 + c_2\right)}_{\text{consumption cost}} + \underbrace{w\left(l_1 + l_2\right)}_{\text{opportunity cost of leisure}}$$

L1 - The agricultural household and the separation property

- The household **produces** $F(L, A)$ and **consumes**; with complete markets it trades labour, land and output at market prices w, r, p.
- Budget constraint:

$$\underbrace{pF(L,A) - wL - rA}_{\Pi\text{: profits of the } \textit{firm}} + \underbrace{w\left(E_1^L + E_2^L\right) + rE^A}_{\text{value of labour \& land endowments}} \geq \underbrace{p\left(c_1 + c_2\right)}_{\text{consumption cost}} + \underbrace{w\left(l_1 + l_2\right)}_{\text{opportunity cost of leisure}}$$

- FOCs $pF_L = w$, $pF_A = r \Rightarrow$ production depends only on $F(\cdot)$ and prices w and r, **not on preferences or endowments**. Recursive: maximise Π first, then utility.

L1 - The agricultural household and the separation property

- The household **produces** $F(L, A)$ and **consumes**; with complete markets it trades labour, land and output at market prices w, r, p.
- Budget constraint:

$$\underbrace{pF(L,A) - wL - rA}_{\Pi\text{: profits of the } \textit{firm}} + \underbrace{w\left(E_1^L + E_2^L\right) + rE^A}_{\text{value of labour \& land endowments}} \geq \underbrace{p\left(c_1 + c_2\right)}_{\text{consumption cost}} + \underbrace{w\left(l_1 + l_2\right)}_{\text{opportunity cost of leisure}}$$

- FOCs $pF_L = w$, $pF_A = r \Rightarrow$ production depends only on $F(\cdot)$ and prices w and r, **not on preferences or endowments**. Recursive: maximise Π first, then utility.
- One missing market (land) is not enough to break this. Several are (e.g., no land market *and* a binding cap on off-farm work): labour use then depends on preferences and endowments.

Evidence: LaFave and Thomas (2016), Central Java

If markets are complete, production does not depend on labour endowment. Here: Household composition affects production $\Rightarrow$ markets are incomplete.

L1 - The unitary model and income pooling

- Members maximise utility **as if** the household were a single decision unit.
- Justifications: identical preferences; a "consensus" welfare function; one member decides (benevolent dictator).
- Key implication - **income pooling**: demands depend on total income, not on who earns it.

Evidence: Duflo (2003), South Africa

Old-age pension received by a **woman** $\Rightarrow$ girls' weight-for-height improves, little effect on boys. Received by a **man** $\Rightarrow$ no effect. Who receives income matters $\Rightarrow$ **Income pooling is rejected**.

- Two members a, b; public goods Q, private goods q^a, q^b; budget $P'Q + p'q^a + p'q^b \leq x$.
- Cooperative models assume **stability** and **Pareto efficiency** (non-cooperative ones do not).
- Every efficient allocation solves

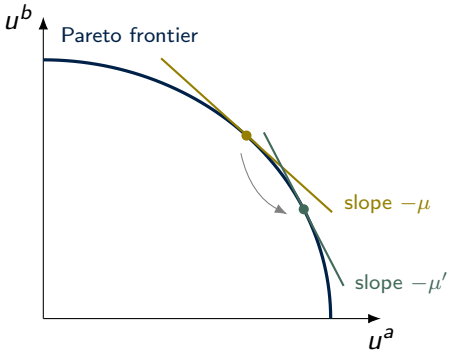
$$\max_{Q,\, q^a,\, q^b} \ \mu(P, p, x, z)\ u^a(Q, q^a, q^b) + u^b(Q, q^a, q^b) \quad \text{s.t. the budget}$$

- μ is the **Pareto weight**: member a's decision power.
- **Distribution factors** z shift μ without changing preferences or the budget set: income shares Y^a/Y, divorce and inheritance laws, (re)marriage markets, outside options.

Why this matters for PS1

μ fixed $\Rightarrow$ back to the unitary model. $\mu = \mu(P, p, x, z) \Rightarrow$ prices and income have substitution, income **and bargaining power** effects; z acts only through power.

The Pareto weight, graphically



- Each μ picks one point on the frontier: where it is tangent to $\mu u^a + u^b = \text{const}$.
- Higher weight $\mu' > \mu \Rightarrow$ the allocation slides towards a.
- Efficiency $\neq$ no conflict: members rationally try to gain power (bargaining, contracts, norms).
- Efficiency $\neq$ fairness: points near the axes are efficient but very unequal.

L2 - Measuring and shifting intrahousehold power

- **Calvi (2020):** Does women's intrahousehold bargaining power over household resources fall as they age, helping explain the "missing" *older* women?
  A. Reduced form: Hindu Succession Act amendments equalised inheritance rights ⇒ better outside options.
  B. Structural: resource shares estimated from an *assignable* good (clothing) via Engel curves.
- **Field et al. (2021):** MGNREGS wages paid into women's *own* bank accounts (+ training): control over income changes, not its amount ⇒ women work more, in the programme and the private sector. The opposite of the basic efficient-bargaining prediction; a **norms** channel rationalises it.
- **Jayachandran, Biradavolu and Cooper (2023):** A five-question survey module on women's agency, validated against qualitative interviews.

L3 - Is the household efficient?

- Cooperative model + production $\Rightarrow$ first maximise household income, then split it by μ.
- So separation now holds *within* the household: inputs go wherever they are most productive, whoever controls the plot.

Udry (1996), Burkina Faso

Households farm many plots, controlled by different members.

- **Prediction:** similar plots (size, quality, crop, year) have the **same yields**.
- **Finding:** plots controlled by women yield less; labour and fertiliser are concentrated on men's plots.
- $\Rightarrow$ **Pareto efficiency is rejected.**

L3 - Why might efficiency fail?

- **Commitment** - Lundberg and Pollak (2003): decisions happen in stages and future power cannot be contracted on (next slide).
- **Exit options** - Walther (2018), Malawi: in patrilineal households men farm 1.5 h/week more, although wage work would raise household income. Men keep the land on divorce, so farming it builds their bargaining power.
- **Private information and actions** - Ashraf, Field and Lee (2014), Zambia: contraception vouchers given to women alone $\Rightarrow$ higher take-up of concealable methods than when given to the couple; welfare effects inconclusive.

L3 - A worked example: power that moves with income

Wife w, husband h, log utilities, and a weight that depends on the husband's income share $s_h = y_h/(y_w + y_h)$:

$$\max_{q_w, q_h} \log q_w + \mu(s_h) \log q_h \;\text{ s.t. }\; q_w + q_h \le y_w + y_h$$

- FOCs: $1/q_w = \lambda, \quad \mu/q_h = \lambda \;\Rightarrow\; q_h = \mu\, q_w$
- $q_w^\star = \dfrac{y_w + y_h}{1 + \mu(s_h)}$
- $\dfrac{\partial q_w^\star}{\partial y_h} = \dfrac{1 + \mu - \mu'(s_h)\, y_w/(y_w + y_h)}{(1+\mu)^2}$

Intuition

A rise in y_h has an **income effect** $(+)$ and a **power effect** $(-)$. If the power effect dominates, the wife vetoes a move that raises total income.

- Fixes: a contract or a compensating transfer — both hard to make credible.
- **Same logic as PS1 (f).**

What children are, economically

- a source of joy (consumption good);
- another consumer;
- another **earner** $\rightarrow$ child labour, PS1 (g)–(j);
- an investment / old-age support;
- a cause of mother wage penalties.

What the models predict

- Parents invest until marginal benefit = marginal cost.
- Heterogeneity *between* households (income, prices, technology) and *within* them (child's gender).
- Unitary or collective? Are kids a normal good? A public good?

L4 - Qian (2008): missing women and the price of tea

- Post-Mao reforms (1978–80) raised returns to **tea** (picked mainly by women) and **orchards** (men).
- Difference-in-differences: tea-suitable vs. other areas, before vs. after. Higher female income $\Rightarrow$ better survival and education of girls.

	Model	Empirical implication
H1	Unitary, kids as investment	symmetric: boys in orchard areas, girls in tea areas
H2	Unitary, girls as luxury goods	*any* income rise balances the sex ratio
H3	Collective, mothers' power	female income raises schooling of all kids, girls most

Magnitude — keep it in mind for PS1 (d)

+20% household income, all given to women, would have brought China's sex imbalance to Western European levels.

PS1 Q1 (a)-(d): the unitary household

Q1 - setup

- A household made of two representative groups: **adults** (consume x_a at price p_a) and **kids** (consume x_k at price p_k).
- Household utility $u_h(x_a, x_k)$ neoclassical: increasing, concave, twice differentiable.
- Price-taking household with exogenous income y_0 (adults' earnings, inelastic labour supply).

Roadmap

(a)-(d)	unitary model, then Cobb–Douglas with weight α
(e)-(f)	collective model with Pareto weight μ
(g)-(j)	back to unitary: child labour h_k and cash transfers

Q1 (a) - the unitary household problem

Question. Under a unitary household model, write down the household's optimisation problem, including the budget constraint.

Q1 (a) - the unitary household problem

Question. Under a unitary household model, write down the household's optimisation problem, including the budget constraint.

$$\max_{x_a,\, x_k} \; u_h(x_a, x_k) \qquad \text{subject to} \qquad y_0 \;\geq\; p_a x_a + p_k x_k$$

- One utility function for the whole household, over what each group consumes. Non-satiation (u_h is increasing), so "=" is fine too.

Q1 (b) - can we defend the unitary assumption here?

Question. The unitary model is often called unrealistic. Which arguments could support its assumptions in this problem?

Q1 (b) - can we defend the unitary assumption here?

Question. The unitary model is often called unrealistic. Which arguments could support its assumptions in this problem?

- Core assumption: utility is maximised **as if** the household were a single decision unit.
- This question is about **adults vs. kids**, not spouses: adults choose the kids' basket.
- **Benevolent dictator:** adults know the kids' preferences and internalise their wellbeing (Becker 1974).

  Much harder to defend *between adults*, where income pooling is rejected (Duflo 2003).

Q1 (b) - is the dictator benevolent? Altruism vs paternalism

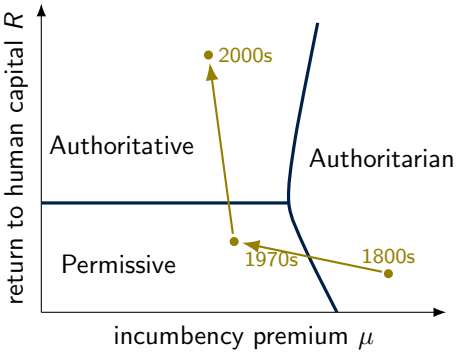
Doepke and Zilibotti (2017): a parent values the child's experience as

$$w(X,a) = \mathbb{E}'_s\Big[\underbrace{(1-\lambda)\,U^y(c^y \mid a)}_{\text{altruism (Becker)}} + \underbrace{\lambda\,U^o(c^y \mid a)}_{\text{paternalism}} + \beta\, v^o(a,h',s')\Big]$$

where $v^o(a,h',s')$ is the parent's value function.

- U^y: the child's own utility function. U^o: the *parent's* own utility function applied to the child's choices.
- The parent can let the child choose (**permissive**), restrict the child's choice set X (**authoritarian**) or mould preferences a (**authoritative**).
- $\lambda = 0$ is the benevolent dictator: never acts against the child's own utility (permissive or neglecting).
- $\lambda = 1$ is the paternalistic dictator: always acts against the child's own utility (authoritarian or authoritative).

Q1 (b) - parenting style depends on the economic environment



Schematic of Doepke and Zilibotti (2017), Fig. 5.

- **Authoritarian** when incumbecy premium μ is high: children do well by following the parent's occupation.
- Otherwise **authoritative** if the return to human capital R is high, **permissive** if it is low.
- History: industrialisation lowers μ; later, R rises.

Q1 (c) - log-linear Cobb-Douglas utility

Question. With $u_h = \alpha \log x_a + (1-\alpha) \log x_k$, $\alpha \in [0,1]$, derive the household demands, x_a and x_k.

- Lagrangian: $\mathcal{L} = \alpha \log x_a + (1-\alpha) \log x_k + \lambda (y_0 - p_a x_a - p_k x_k)$

Q1 (c) - log-linear Cobb-Douglas utility

Question. With $u_h = \alpha \log x_a + (1-\alpha) \log x_k$, $\alpha \in [0,1]$, derive the household demands, x_a and x_k.

- Lagrangian: $\mathcal{L} = \alpha \log x_a + (1-\alpha) \log x_k + \lambda (y_0 - p_a x_a - p_k x_k)$
- FOCs: $\frac{\alpha}{x_a} = \lambda p_a, \qquad \frac{1-\alpha}{x_k} = \lambda p_k, \qquad p_a x_a + p_k x_k = y_0$

Q1 (c) - log-linear Cobb-Douglas utility

Question. With $u_h = \alpha \log x_a + (1-\alpha) \log x_k$, $\alpha \in [0,1]$, derive the household demands, x_a and x_k.

- Lagrangian: $\mathcal{L} = \alpha \log x_a + (1-\alpha) \log x_k + \lambda (y_0 - p_a x_a - p_k x_k)$
- FOCs: $\dfrac{\alpha}{x_a} = \lambda p_a, \qquad \dfrac{1-\alpha}{x_k} = \lambda p_k, \qquad p_a x_a + p_k x_k = y_0$
- Rearrange: $p_a x_a = \alpha/\lambda$, $p_k x_k = (1-\alpha)/\lambda$; sum $\Rightarrow \lambda = 1/y_0$ ▸ λ as marginal utility of income

Q1 (c) - log-linear Cobb-Douglas utility

Question. With $u_h = \alpha \log x_a + (1-\alpha) \log x_k$, $\alpha \in [0,1]$, derive the household demands, x_a and x_k.

- Lagrangian: $\mathcal{L} = \alpha \log x_a + (1-\alpha) \log x_k + \lambda (y_0 - p_a x_a - p_k x_k)$
- FOCs: $\frac{\alpha}{x_a} = \lambda p_a, \qquad \frac{1-\alpha}{x_k} = \lambda p_k, \qquad p_a x_a + p_k x_k = y_0$
- Rearrange: $p_a x_a = \alpha/\lambda$, $p_k x_k = (1-\alpha)/\lambda$; sum $\Rightarrow \lambda = 1/y_0$ ▸ λ as marginal utility of income
- Substitute Lagrande multiplier back into FOCs.

Q1 (c) - log-linear Cobb-Douglas utility

Question. With $u_h = \alpha \log x_a + (1-\alpha) \log x_k$, $\alpha \in [0,1]$, derive the household demands, x_a and x_k.

- Lagrangian: $\mathcal{L} = \alpha \log x_a + (1-\alpha) \log x_k + \lambda (y_0 - p_a x_a - p_k x_k)$
- FOCs: $\dfrac{\alpha}{x_a} = \lambda p_a, \qquad \dfrac{1-\alpha}{x_k} = \lambda p_k, \qquad p_a x_a + p_k x_k = y_0$
- Rearrange: $p_a x_a = \alpha/\lambda$, $p_k x_k = (1-\alpha)/\lambda$; sum $\Rightarrow \lambda = 1/y_0$ ▸ λ as marginal utility of income
- Substitute Lagrande multiplier back into FOCs.
- Marshallian Demands:

$$\boxed{x_a^\star = \frac{\alpha}{p_a} y_0, \qquad x_k^\star = \frac{1-\alpha}{p_k} y_0}$$

Q1 (c) - log-linear Cobb-Douglas utility

Question. With $u_h = \alpha \log x_a + (1-\alpha) \log x_k$, $\alpha \in [0,1]$, derive the household demands, x_a and x_k.

- Lagrangian: $\mathcal{L} = \alpha \log x_a + (1-\alpha) \log x_k + \lambda (y_0 - p_a x_a - p_k x_k)$
- FOCs: $\dfrac{\alpha}{x_a} = \lambda p_a, \qquad \dfrac{1-\alpha}{x_k} = \lambda p_k, \qquad p_a x_a + p_k x_k = y_0$
- Rearrange: $p_a x_a = \alpha/\lambda$, $p_k x_k = (1-\alpha)/\lambda$; sum $\Rightarrow \lambda = 1/y_0$ ▸ λ as marginal utility of income
- Substitute Lagrande multiplier back into FOCs.
- Marshallian Demands: $\boxed{x_a^\star = \dfrac{\alpha}{p_a} y_0, \qquad x_k^\star = \dfrac{1-\alpha}{p_k} y_0}$
- Constant expenditure shares $\dfrac{p_a x_a^\star}{y_0} = \alpha$, $\dfrac{p_k x_k^\star}{y_0} = 1-\alpha$.

Q1 (d) - a 20% income increase

Question. Income rises by 20% to $y_1 = 1.2\, y_0$. How does the allocation change? Who benefits most, adults or kids?

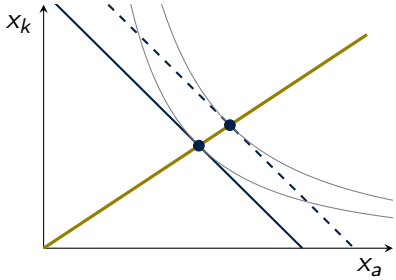
Q1 (d) - a 20% income increase

Question. Income rises by 20% to $y_1 = 1.2\, y_0$. How does the allocation change? Who benefits most, adults or kids?

$$\tilde{x}_a = \frac{\alpha}{p_a}\, 1.2\, y_0 = 1.2\, x_a^\star$$

$$\tilde{x}_k = \frac{1-\alpha}{p_k}\, 1.2\, y_0 = 1.2\, x_k^\star$$

- Shares stay at α, $1-\alpha$; income elasticity $= 1$.
- Expansion path: a ray from the origin (homothetic preferences).



$\alpha = 0.6$, $p_a = p_k = 1$, $y_0 = 10$, $y_1 = 12$

Q1 (d) - so who benefits most?

- **Proportionally, both equally:** each group's consumption rises by 20%, and $\log(1.2\,x^\star) = \log 1.2 + \log x^\star$, the same gain ($\approx 0.18$) for both.
- **In levels**, extra spending is $0.2\,\alpha\,y_0$ for adults vs. $0.2\,(1-\alpha)\,y_0$ for kids: larger for the group with the larger share.

Common pitfall

"Equal benefit" does not mean the budget is split equally: $p_a x_a / p_k x_k = \alpha/(1-\alpha)$, equal only if $\alpha = \frac{1}{2}$.

Takeaway

Here it is irrelevant *who* receives the extra 20% (income pooling). Contrast with Qian (2008): 20% more income, given to women alone, would help closing China's sex imbalance.

PS1 Q1 (e)-(f): the collective household

Q1 (e) - interpreting μ

Question. With $u_h = \mu \log x_a + (1-\mu) \log x_k$: interpret μ. When is it (not) equivalent to α?

Q1 (e) - interpreting μ

Question. With $u_h = \mu \log x_a + (1-\mu) \log x_k$: interpret μ. When is it (not) equivalent to α?

	α (unitary)	μ (collective)
What it is	taste parameter of one household utility	**Pareto weight** on members' utilities $u_a = \log x_a$, $u_k = \log x_k$
Meaning	how much the household values each good	**Bargaining power**, e.g. from outside options (plausibly higher for adults)
Depends on	nothing: fixed	potentially $\mu(p, y, z)$: prices, income, other distribution factors like who holds the income (Duflo, 2003)

- **Equivalent** if μ is exogenous and fixed: same problem, so $x_a^\star = \mu y / p_a$, $x_k^\star = (1-\mu) y / p_k$.
- **Not equivalent** once μ moves with prices, income or z.

Q1 (e) - for discussion: do kids bargain?

- Children have essentially no outside options, so "their" weight $1 - \mu$ is unlikely to come from their own bargaining.
- A more natural reading: $1 - \mu$ reflects the power of the **adult who cares most about the kids**.
- This is the channel in the lecture evidence:
  - Duflo (2003): pensions to grandmothers, not grandfathers, improve girls' nutrition;
  - Qian (2008), H3: higher female income $\Rightarrow$ more investment in children, especially girls.

Intuition

With μ fixed, the collective model is observationally identical to the unitary one. It is testable only when μ responds to something observable.

Q1 (f) - income changes in the collective model

Question. How does an arbitrary income increase affect the allocation in the collective model?

Q1 (f) - income changes in the collective model

Question. How does an arbitrary income increase affect the allocation in the collective model?

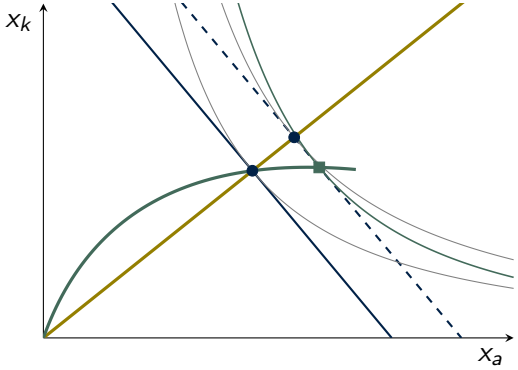
With $\mu = \mu(y)$:

$$x_a^\star = \frac{\mu(y)\, y}{p_a}, \qquad x_k^\star = \frac{(1-\mu(y))\, y}{p_k}$$

$$\Rightarrow \quad \varepsilon_a = 1 + \frac{y\, \mu'(y)}{\mu}, \qquad \varepsilon_k = 1 - \frac{y\, \mu'(y)}{1-\mu}$$

- $\mu'(y) = 0$ (fixed weight): proportional gains, **as in (d)**.
- $\mu'(y) > 0$ (adults earn the extra income): adults gain more than proportionally, kids less; kids even *lose* if $y\mu' > 1 - \mu$.
- $\mu'(y) < 0$ (kids earn the extra income): kids gain more; adults less; adults even *lose* if $-y\mu' > \mu$.

Q1 (f) - the picture: Income Expansion Paths



- Income rises, prices fixed: the budget line shifts out in parallel (slope $-p_a/p_k$ unchanged).
- **Unitary model** path: a ray. In **collective model**, with μ rising in y, the path bends towards x_a.
- Same income gain, but μ shifts with it $\Rightarrow$ different allocation.
- *Question: Income in the initial equilibrium?*

PS1 Q1 (g)-(j): child labour under unitary household model

Q1 (g)-(j) - new setup

- Back to the **unitary** model; one consumption bundle x with price normalised to 1.
- Kids split their time: work h_k and play/study $l_k = 1 - h_k$, with $h_k \in [0, 1]$.
- Child work pays the market wage w_k; earnings add to baseline income y_0.

$$u_h(x, h_k) = \log x + \theta\,(1 - h_k), \qquad x \le y_0 + w_k h_k$$

Intuition

Utility is **quasi-linear** in the kids' time: each unit of play/study is worth a constant θ, while consumption has diminishing marginal utility.

Q1 (g) - interpreting θ

Question. Interpret θ. Do you expect it to be positive or negative? Why might it differ across families or contexts?

Q1 (g) - interpreting θ

Question. Interpret θ. Do you expect it to be positive or negative? Why might it differ across families or contexts?

- θ is the (constant) marginal value the household places on the kids' **non-work time**, since $1 - h_k = l_k$.
- Expected **positive**: play and study raise the kids' wellbeing and human capital.
- Why it varies:
  - social norms on child work and child rights;
  - local institutions, formal and informal (laws, enforcement);
  - returns to schooling, and the availability and quality of schools;
  - parental altruism vs. paternalism (Doepke and Zilibotti 2017): $\theta = (1 - \lambda)\,\theta_k + \lambda\,\theta_p$ mixes the kid's own value of play/study (θ_k) with the parents' view (θ_p); the benevolent dictator ($\lambda = 0$) has $\theta = \theta_k$.

Q1 (h) - solving for child labour

Question. Write the household problem and solve for $h_k^\star$ as a function of exogenous variables.

- Non-satiation $\Rightarrow x = y_0 + w_k h_k$. Substitute: $\max_{h_k \in [0,1]} \log(y_0 + w_k h_k) + \theta\,(1 - h_k)$

Q1 (h) - solving for child labour

Question. Write the household problem and solve for $h_k^\star$ as a function of exogenous variables.

- Non-satiation $\Rightarrow x = y_0 + w_k h_k$. Substitute: $\max_{h_k \in [0,1]} \log(y_0 + w_k h_k) + \theta(1 - h_k)$
- FOC: $\dfrac{w_k}{y_0 + w_k h_k} - \theta = 0$ SOC: $-\dfrac{w_k^2}{(y_0 + w_k h_k)^2} < 0 \Rightarrow$ global maximum

Q1 (h) - solving for child labour

Question. Write the household problem and solve for $h_k^\star$ as a function of exogenous variables.

- Non-satiation $\Rightarrow x = y_0 + w_k h_k$. Substitute: $\max_{h_k \in [0,1]} \log(y_0 + w_k h_k) + \theta\,(1 - h_k)$
- FOC: $\dfrac{w_k}{y_0 + w_k h_k} - \theta = 0$ SOC: $-\dfrac{w_k^2}{(y_0 + w_k h_k)^2} < 0 \Rightarrow$ global maximum
- Interior solution: $\boxed{h_k^\star = \dfrac{w_k - \theta y_0}{\theta\, w_k} = \dfrac{1}{\theta} - \dfrac{y_0}{w_k}}$

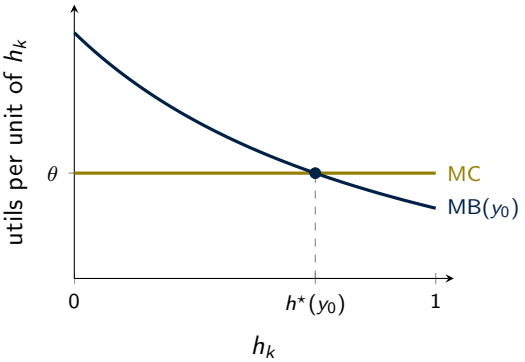
Q1 (h) - solving for child labour

Question. Write the household problem and solve for $h_k^\star$ as a function of exogenous variables.

- Non-satiation $\Rightarrow x = y_0 + w_k h_k$. Substitute: $\max_{h_k \in [0,1]} \log(y_0 + w_k h_k) + \theta\,(1 - h_k)$
- FOC: $\dfrac{w_k}{y_0 + w_k h_k} - \theta = 0$ SOC: $-\dfrac{w_k^2}{(y_0 + w_k h_k)^2} < 0 \;\Rightarrow$ global maximum
- Interior solution: $\boxed{h_k^\star = \dfrac{w_k - \theta y_0}{\theta\, w_k} = \dfrac{1}{\theta} - \dfrac{y_0}{w_k}}$
- Comparative statics (*interpretation?*):

$$\partial h_k^\star / \partial y_0 = -1/w_k < 0, \quad \partial h_k^\star / \partial w_k = y_0 / w_k^2 > 0, \quad \partial h_k^\star / \partial \theta = -1/\theta^2 < 0$$

Q1 (h) - FOC in picture: marginal benefit = marginal cost



$w_k = 1$, $\theta = 0.6$, $y_0 = 1 \Rightarrow h_k^\star = 2/3$

- **MB** of an extra unit of child work: wage $\times$ marginal utility of consumption,

$$\text{MB} = w_k \cdot \frac{1}{y_0 + w_k h_k},$$

  falling in h_k.
- **MC**: forgone play/study, constant at θ.

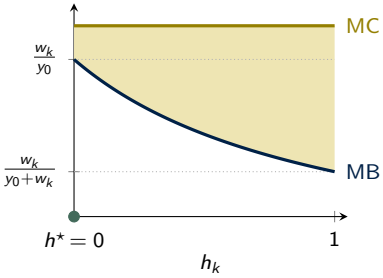
Q1 (i) - when is there child labour?

Question. Under which conditions do we observe strictly positive child labour? Interpret.

Q1 (i) - when is there child labour?

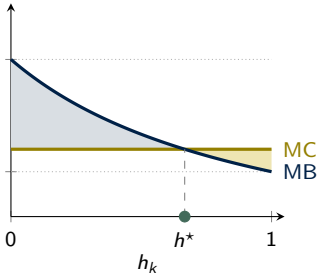
Question. Under which conditions do we observe strictly positive child labour? Interpret.

No child labour



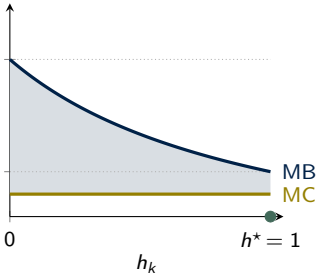
$\theta \geq w_k/y_0$
MC $>$ MB at every h_k

Some child labour



$w_k/(y_0 + w_k) < \theta < w_k/y_0$
MB $=$ MC inside $[0, 1]$

Full-time child labour



$\theta \leq w_k/(y_0 + w_k)$
MB $>$ MC at every h_k

Q1 (j) - which policies raise y_0?

Question. A social policy raises baseline income from y_0 to y_1, all else constant. Which real-world policies fit this description?

Q1 (j) - which policies raise y_0?

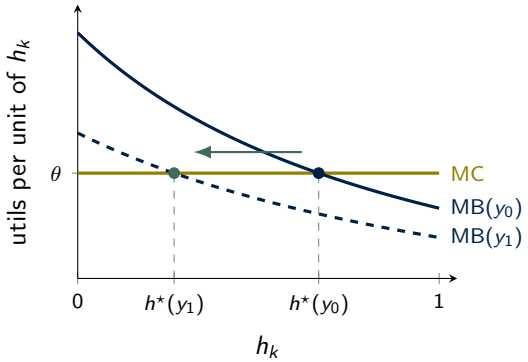
Question. A social policy raises baseline income from y_0 to y_1, all else constant. Which real-world policies fit this description?

- **Unconditional cash transfers.**
- **Social pensions**, e.g. the South African old-age pension in Duflo (2003).
- Higher **adult earnings**: productivity gains, lower taxes, ...

Common pitfall

Conditional cash transfers (paid only if kids attend school) do *not* fit: the condition changes the budget set itself - in effect the price of child time - not just y_0.

Q1 (j) - effect on child labour



$y_1 = 1.4 \Rightarrow h_k^\star$ falls from 2/3 to 4/15

- Higher y_0 lowers the MB curve, so $h_k^\star$ falls: $\partial h_k^\star / \partial y_0 = -1/w_k$.
- Once $y_1 \geq w_k/\theta$: corner, $h_k^\star = 0$.
- If θ varies across households, the *share* with child labour, $\Pr(\theta < w_k/y)$, falls with y.

Wrap-up

What to take away from PS1

- **Unitary + Cobb-Douglas:** shares fixed by α; income changes scale everything proportionally; who earns is irrelevant (income pooling).
- **Collective:** same algebra, different meaning. μ is intra-household bargaining power; once it moves with *who* earns the income, income pooling fails.
- **Child labour:** watch the corners. $h_k^\star > 0$ iff $y_0 < w_k/\theta$: a poverty threshold.
- **Policy:** unconditional transfers act through income alone; conditional ones also change the price of child time.

Takeaway

With a fixed μ, the collective model is observationally equivalent to the unitary one. Once μ depends on *who* holds the money, income pooling fails — which is what Duflo (2003) and Qian (2008) find.

Appendix: λ as marginal utility of income

- Define the indirect utility function $v(p, y) = \max_x u(x)$ s.t. $p \cdot x \leq y$.
- $v(p, y) = u(x^*(p, y))$ where $x^*(p, y)$ is the Marshallian demand.
- The change in utility from a change in income is $D_y v(p, y) = \nabla_x u(x^*(p, y)) \cdot D_y x^*(p, y)$.
- From the FOC $\implies \nabla_x u(x^*(p, y)) = \lambda p$ (Kuhn-Tucker conditions), so $D_y v(p, y) = \lambda p \cdot D_y x^*(p, y)$.
- But then note that $p \cdot D_y x^*(p, y) = 1$ (the budget constraint is binding), so $D_y v(p, y) = \lambda$.

◂ Back to Q1 (c)